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Ask a question

Dynamic query input

What is Effect of Cladding Material?

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Get answer

Answer summary

Summary The effect of cladding material on the performance of a heterogeneous slot waveguide is studied, showing that the choice of cladding material affects the overlap factor and electric field distribution.

The refractive index of the upper cladding material influences the modal behavior, with SiO2 cladding exhibiting a greater confining effect on the optical field at 1530 nm.

Key Points

- The choice of cladding material affects the overlap factor and electric field distribution [1].
- SiO2 cladding confines a greater area of the optical field to the slot waveguide compared to air cladding at 1530 nm [1].
- The overlap factor reaches its maximum for the cladding index of 1.56 at 980 nm, whereas it exhibits a cliff drop with cladding index above 1.43 at 1530 nm [2].
- The refractive index difference between the waveguide cross sections with air-clad and SiO2-clad is large, resulting in a symmetrical refractive index distribution of the SiO2-clad amplifier [3].

References

[1] High-performance_amplifier_design_.pdf, page 5
[2] High-performance_amplifier_design_.pdf, page 4
[3] High-performance_amplifier_design_.pdf, page 4

Documents used

- High-performance_amplifier_design_.pdf

Citations

[1] High-performance_amplifier_design_.pdf, page 5, chunk 124

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(d) and SiO2-clad (e). the cladding material and the active gain medium. Moreover, the insets in Fig. 4 show the electric field distributions for the upper cladding materials of air and SiO 2, respectively, at the corresponding wavelengths. On the whole, the change in cladding material exhibits a hig

[2] High-performance_amplifier_design_.pdf, page 4, chunk 122

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overlap factor reaches its maximum for the cladding index of 1.56 at 980 nm, whereas the overlap factor exhibits a cliff drop with cladding index above 1.43 at 1530 nm. This is possibly attributed to the reduced refractive index differenceΔn between

[3] High-performance_amplifier_design_.pdf, page 4, chunk 121

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waveguide cross sections with air-clad (left) and SiO2-clad (right). The overlap factor versus cladding index from 1 to 1.6 with Wslot = 150 nm, HSiN = 400 nm, and WSiN = 300 nm at λ = 980 nm (b) and λ = 1530 nm (c). Insets show the normalized TE mode profiles with air and SiO 2 as upper cladding mat

[4] **High-performance_amplifier_design_.pdf**, page 4, chunk 120

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a mode overlap factor as possible with a small aspect ratio and effective mode area. We chose H SiN = 400 nm and WSiN = 300 nm as the optimized device parameters for the paperclip slot waveguide. C. Effect of Cladding Material Refractive index along the z-direction of waveguide cross sections with a